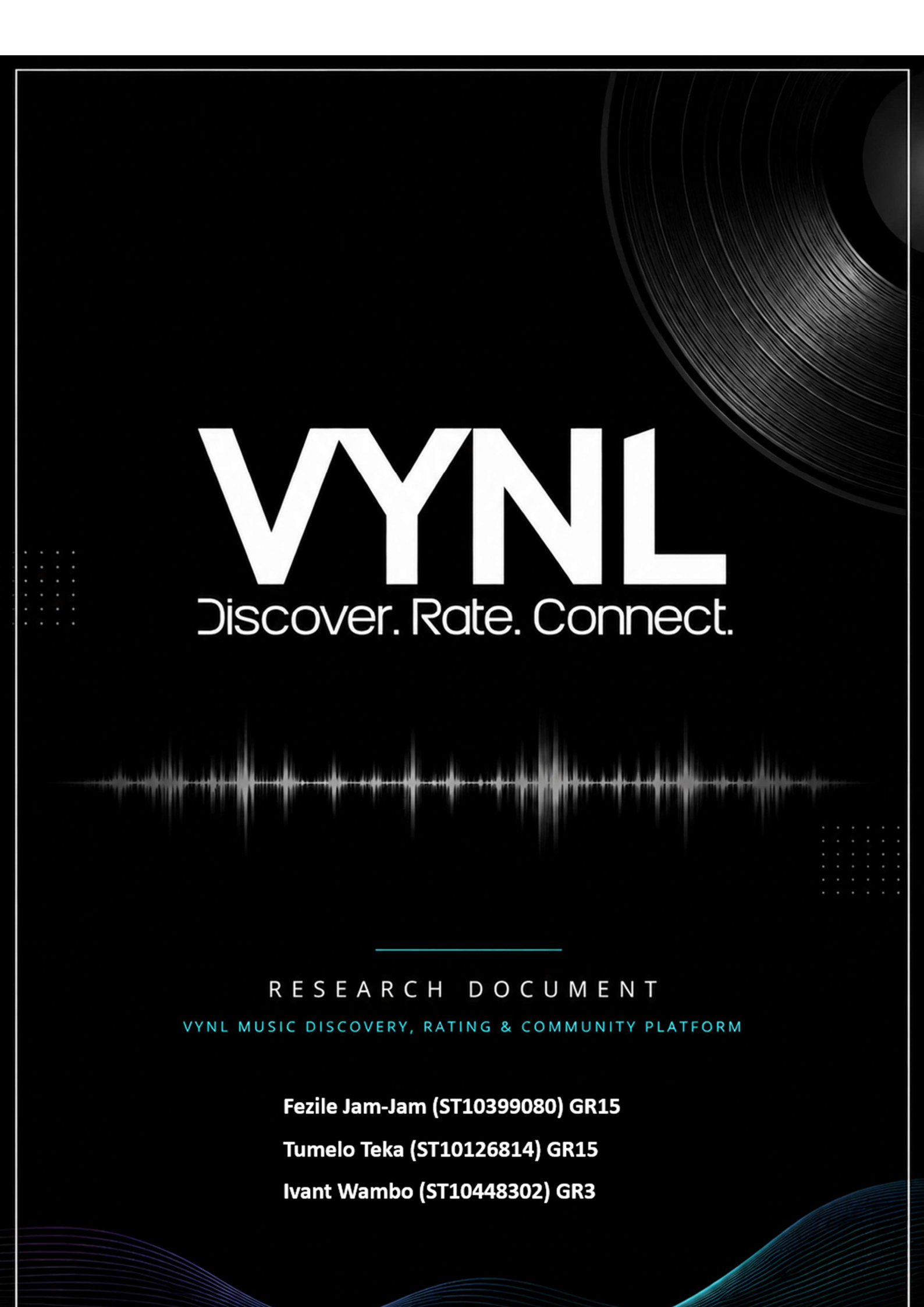




VYNL

Discover. Rate. Connect.

RESEARCH DOCUMENT

VYNL MUSIC DISCOVERY, RATING & COMMUNITY PLATFORM

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Introduction

The Music Streaming Industry Has Significantly Evolved Over the Past Decade, With Users Increasingly Seeking Platforms That Offer More Than Just Music Playback. Modern Music Applications Provide Features Such as Album and Song Tracking, Personalized Recommendations, Community Reviews, Music Discovery, and The Ability to Organize Listening Experiences. The Purpose of This Research Is to Analyze Existing Android Music Applications to Identify Their Strengths, Weaknesses and Innovative Features That Can Be Incorporated into The Development of Our Album and Song Discovery, Tracking and Recommendation Application.

This Report Examines Three Established Music Applications; Last.fm, Musicboard and Rate Your Music, To Understand How They Approach Music Discovery, Album and Song Organization, User Interaction and Recommendation Systems. Each Application Will Be Evaluated Based on Its Overall Functionality, Key Strengths and Weaknesses and An Analysis of How Its Features May Have Been Implemented Using Android Development Technologies and APIs.

Following The Individual App Analyzes, The Report Includes an Infographic That Provides a Visual Comparison of The Three Applications, Highlighting Their Similarities and Differences. The Report Then Identifies and Justifies Features That Will Be Incorporated into Our Own Application Based on The Findings of u. Finally, The Report Concluded by Summarizing the Key Insights Gained from The Analysis and Explaining How These Finding Will Guide the Planning and Development of Our Album and Song Discovery, Tracking and Recommendation Application.

Overview Of Last.fm



Figure 1: Last.fm Logo

Source: Last.fm (2026a).

Last.fm is a music discovery and listening analytics platform launched in 2002. Its primary feature is scrobbling (automatic recording of every song a user listens to from supported services such as Spotify, Apple Music, YouTube Music, Tidal and local music players) (Last.fm, 2026a; Last.fm, 2026b). The platform builds detailed listening profiles, generates personalized statistics and recommends music based on user preferences (Last.fm, 2026b).

Strengths And Weaknesses of Last.fm

Strengths	Weaknesses
Large Music Database: Community-Generated Metadata Improves Artist Information, Tags and Recommendations (Last.fm, 2026c; Last.fm, 2026d).	Outdated User Interface: Compared to Modern Social Apps, The Design Appears Old-Fashioned.
Cross-Platform Integration: Supports Spotify, Apple Music, YouTube Music, Tidal and Other Services, Creating A Unified Listening History (Last.fm, 2026a; Last.fm, 2026b).	No Integrated Music Streaming: Users Generally Listen Using Third-Party Streaming Services Instead of Directly Inside Last.fm (Last.fm, 2026a; Last.fm, 2026b).
Detailed Listening Statistics: Provides Lifetime, Yearly, Monthly and Weekly listening Reports, Allowing Users to Analyze Music Habits Using Data Visualization (Last.fm, 2026a; Last.fm, 2026b).	Dependence On External Services: If Spotify or Another Service Changes Its API, Tracking Functionality May Be Affected.
Personalized Recommendations: Uses Collaborative Filtering and Listening History to Recommend Artists, Albums and Tracks Suited to The User's Tastes (Last.fm, 2026b; Last.fm, 2026e).	Limited Social Interaction: Many Community Features Such as Groups and Forums Were Removed Over Time.

Why These Strengths Are Considered Are Innovative

- **Large Music Database:** Provides Access to An Extensive Catalogue of Artists, Albums, Tracks and Genres, Allowing Users to Discover Both Mainstream and Niche Music from A Single Platform (Last.fm, 2026c; Last.fm, 2026d).
- **Cross-Platform Integration:** Combines Listening Activity from Multiple Streaming Services and Music Players into Unified Profile, Giving Users a Complete and Consistent Listening History Regardless of The Platform They Use. (Last.fm, 2026a; Last.fm, 2026b).
- **Detailed Listening Statistics:** Converts Listening History into Meaningful Insights Through Visual Statistics and Trends, Enabling Users to Better Understand and Analyze Their Music Preferences Over Time (Last.fm, 2026a; Last.fm, 2026b).

- **Personalized Recommendations:** Uses Individual Listening Habits and Preferences to Recommend Artists and Albums Tailored to Each User, Providing A More Relevant and Engaging Music Discovery Experience Than Generic Recommendations (Last.fm, 2026; Last.fm, 2026e).

How The App Was Implemented

From My Current Knowledge and Understanding of This App Was Implemented Using the Following Tech Stack:

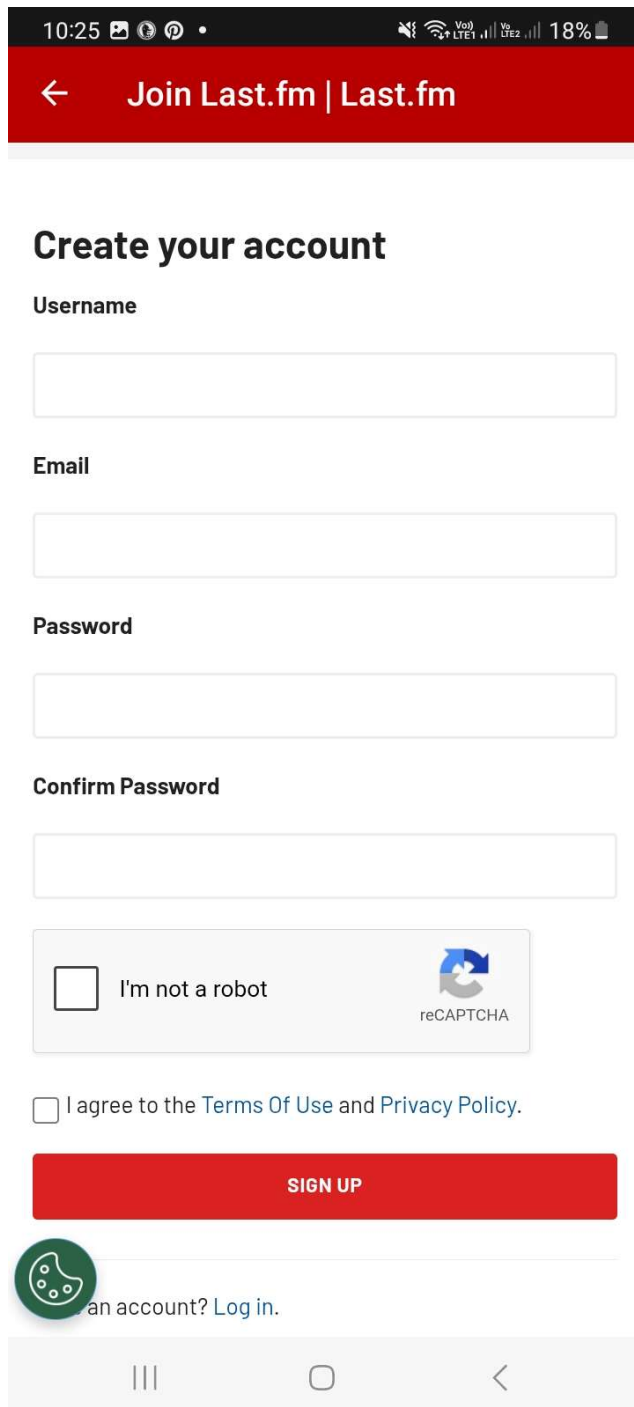
Android Element/Component	Purpose
Kotlin Or Java	Main Programming Language
MVVM Architecture	Separates UI, Business Logic and Data Handling (Android Developers, 2026a; Android Developers, 2026b).
Room Database	Stores Cached Listening History Locally (Android Developers. 2026b).
Retrofit	Connect To Last.fm And Streaming Service APIs (Android Developers, 2026c).
Firebase Authentication	User Login and Authentication
RecyclerView	Displays Lists of Artists, Albums and Tracks
Material Design Components	Modern Android User Interface (Android Developers, 2026d).
WorkManager	Background Synchronization of Listening Data
MPAndroidChart	Creates Listening Statistics Graphs
Coroutines + Flow	Asynchronous API Requests and Live Data Updates (Android Developers, 2026a; Android Developers, 2026e).

The Scrobbling Feature Would Most Probably Work By:

1. Detecting Currently Playing Songs Connected Music Services (Last.fm, 2026f).
2. Sending Track Information to A Backend Server Using REST APIs (Last.fm, 2026g).
3. Storing Play History in A Cloud Database (Last.fm, 2026a; Last.fm, 2026b).
4. Calculating Statistics Using Backend Algorithms (Last.fm, 2026a; Last.fm, 2026h).
5. Returning Recommendations to The Android Application (Last.fm, 2026b).

App Screenshots (Last.fm)

Register Screen



The screenshot shows the 'Join Last.fm | Last.fm' app interface. At the top is a red header bar with a back arrow and the text 'Join Last.fm | Last.fm'. Below this is the title 'Create your account'. The form consists of four input fields: 'Username', 'Email', 'Password', and 'Confirm Password'. Below the 'Password' field is a reCAPTCHA widget with the text 'I'm not a robot' and a checkbox. Below the reCAPTCHA is a checkbox for 'I agree to the Terms Of Use and Privacy Policy.' A red 'SIGN UP' button is positioned below the agreement checkbox. At the bottom, there is a green circular icon with a white 'X' and the text 'Don't have an account? Log in.' The bottom of the screen shows a standard Android navigation bar with three icons: a square, a circle, and a triangle.

10:25 18%

← Join Last.fm | Last.fm

Create your account

Username

Email

Password

Confirm Password

☐ I'm not a robot reCAPTCHA

☐ I agree to the [Terms Of Use](#) and [Privacy Policy](#).

SIGN UP

 Don't have an account? [Log in](#).

Figure 2: Last.fm Register Screen

Source: Last.fm (2026a).

This Is the Register Page for Last.fm, This Is Where the User Will Be Able to Create an Account If They Are Using the Application for The First Time. As Seen Above, In Order to Register They Need to Provide Username, Email, Password, Confirm That They Are Not a Robot and Agree to The Terms of Use and Privacy Policy.

Login Screen

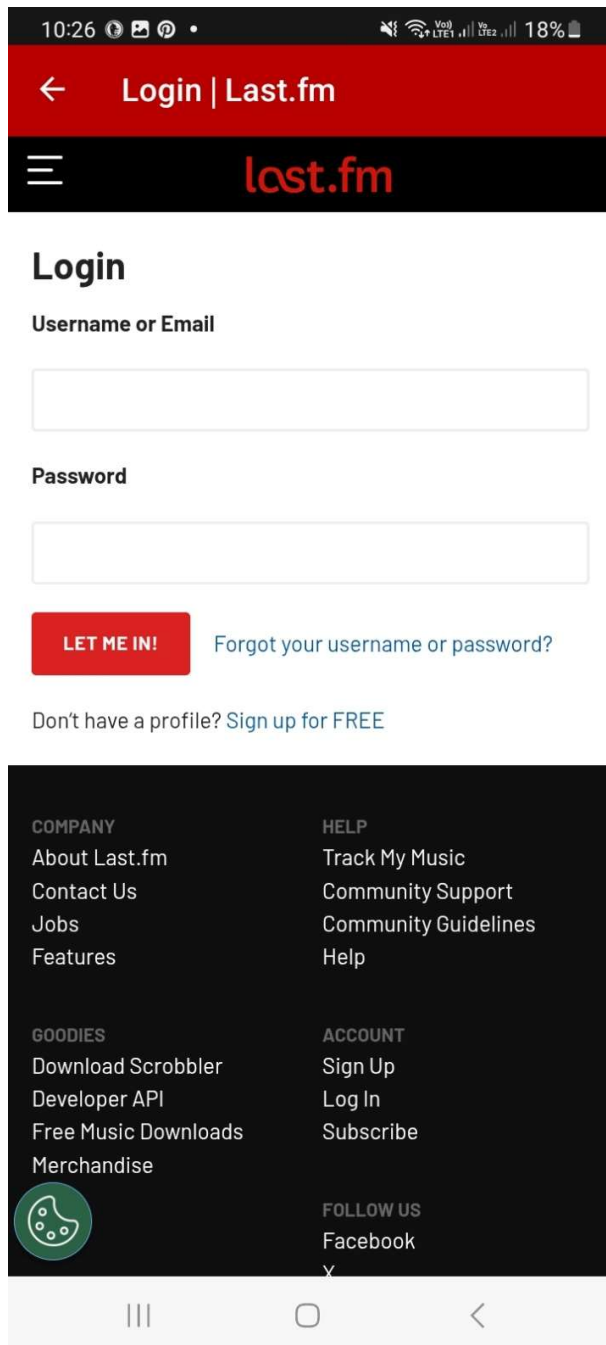


Figure 3: Last.fm Login Screen

Source: Last.fm (2026a).

This Is the Login Page; This Is Where Users Are Able to Sign into The Application Provided That They Have Previously Made an Account on The App. As Seen Above They Can Sign in Using Their Username or Email and The Password Linked to It.

Charts Screen

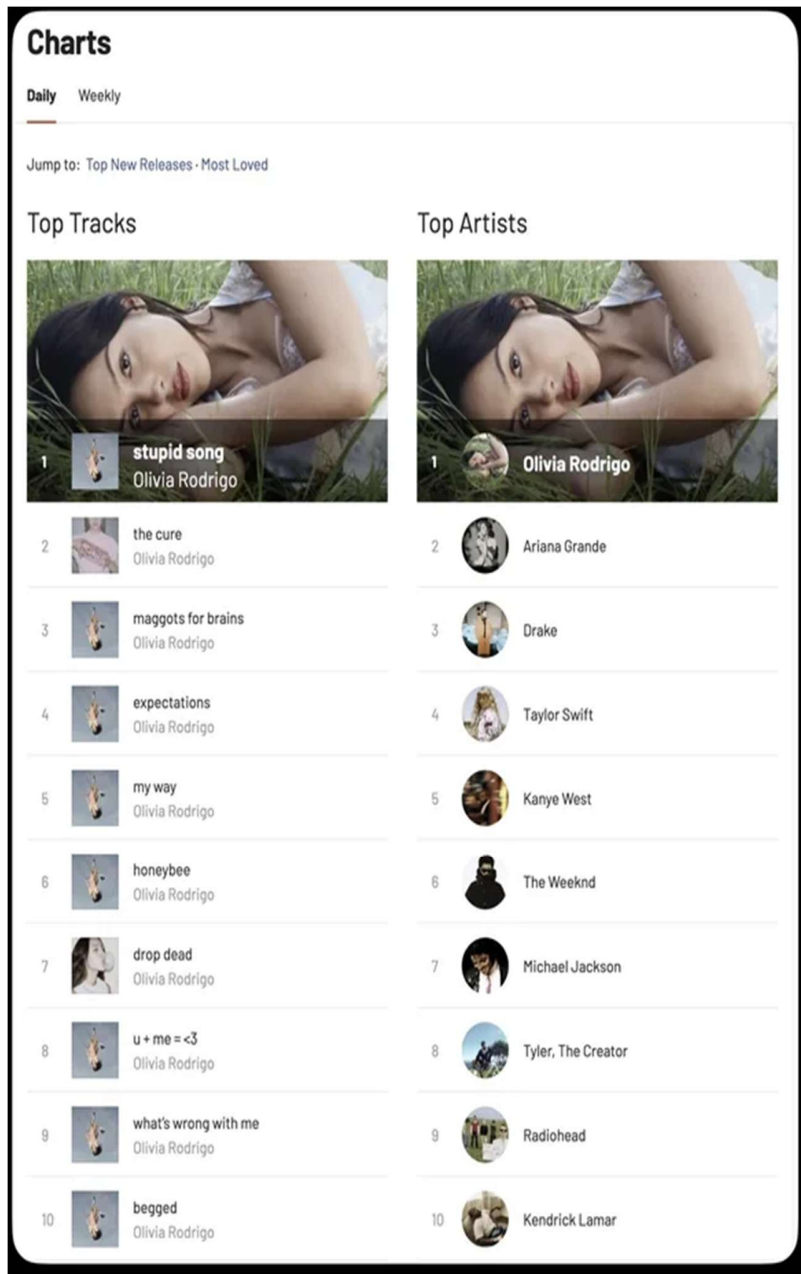


Figure 4: Last.fm Charts Screen

Source: Last.fm (2026a).

This Is the Charts Page, Here the User Is Able to See Current Global Rankings of Albums and Artists, As Such They Will Be Able to See What Albums or What Artists Are Currently Ranked Globally.

Profile Screen

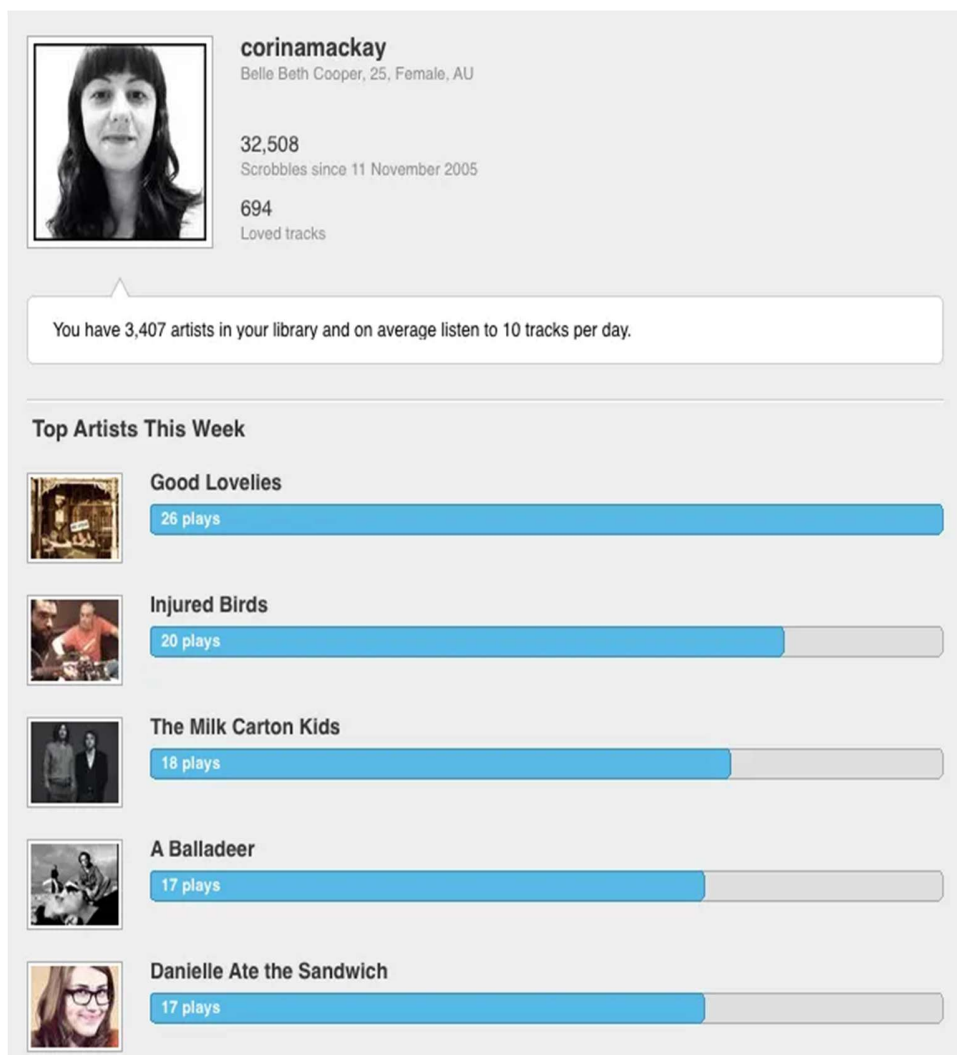


Figure 5: Last.fm Profile Screen

Source: Last.fm (2026a).

This Is the Profile Page, Here the User Is Able to See and Edit Information Regarding Their Profile Such as Their Name, Username, Age, Gender Also Which Artists Are Currently In Their Library And The Average Number Of Songs That They Listen To Daily.

Overview Of Musicboard

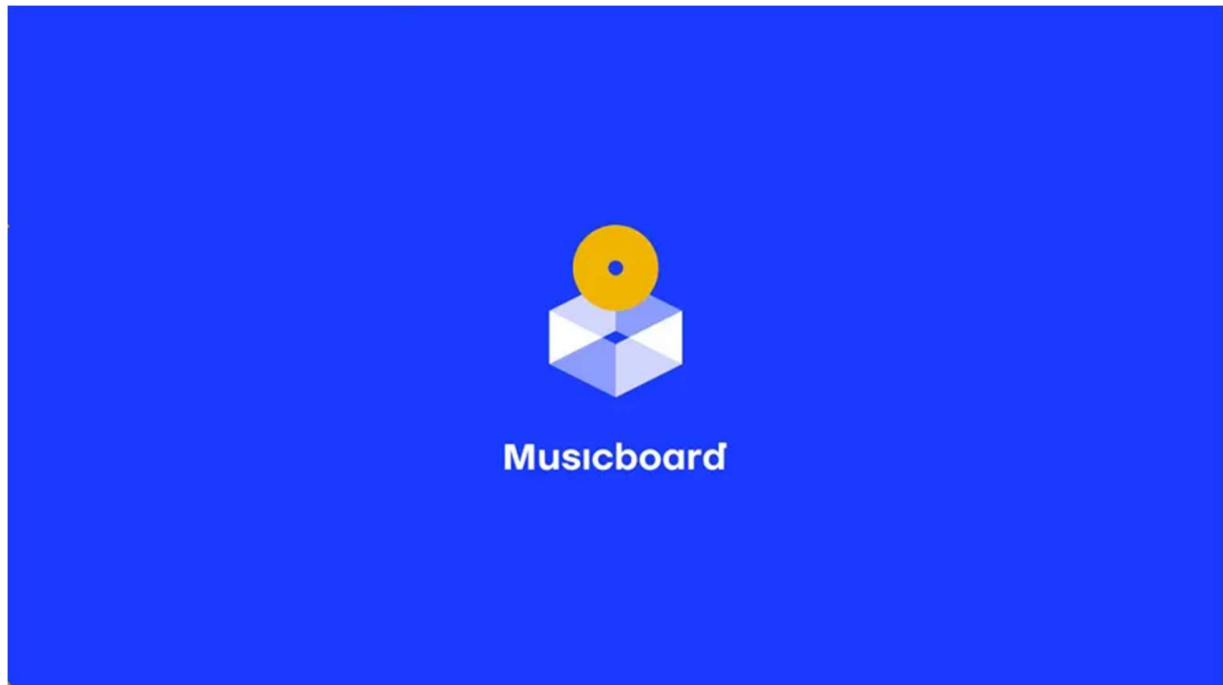


Figure 6: Musicboard Logo

Source: Musicboard (2026a).

Musicboard is often described as the "Letterboxd for music." Rather than automatically tracking listening history, users manually rate, review, and organize albums while interacting with a social community of music enthusiasts. The platform focuses on music criticism, discovery, and discussion (Musicboard, 2026a).

Strengths And Weaknesses of Musicboard

Strengths	Weaknesses
Rating System: Makes It Easy to Compare Albums Through Community Ratings (Musicboard, 2026a).	Smaller User Community: Less Content Compared to Larger Music Platforms Like Spotify or Last.fm
Social Discovery: Users Discover New Music Through Reviews Instead of Algorithms Alone (Musicboard, 2026a; Musicboard, 2026b).	Less Historical Data: Does Not Automatically Create Complete Listening Histories.

Release Tracking: Notifies Users About New Album Release from Favorite Artists (Musicboard, 2026a).	Manual Logging: Users Must Rate or Review Music Themselves Rather Than Automatic Tracking.
Profile Customization: Allow Users to Personalize Their Profiles with Favorite Albums and Statistics (Musicboard, 2026a).	Limited Streaming Integration: Primarily Focuses on Reviews Than Listening (Musicboard, 2026a).

Why These Strengths Are Considered Are Innovative

- Rating System:** Transforms User Opinions into Valuable Community Data, Enabling Personalized Recommendations and Helping Users Quickly Identify Highly Regarded. It Leverages Collective Feedback Rather Than Relying Solely on Algorithms (Musicboard, 2026a).
- Social Discovery:** Enables Users to Discover Music Through Reviews, Ratings and Recommendations from Other Listeners, Creating A More Authentic and Community-Driven Alternative to Algorithm-only Recommendations (Musicboard, 2026a; Musicboard, 2026b).
- Release Tracking:** Automatically Keeps Users Informed About New Releases from Their Favorite Artists, Ensuring They Never Miss New Music and Encouraging Continuous Engagement with The Application (Musicboard, 2026a).
- Profile Customization:** Allows Users to Express Their Musical Identity Through Personalized Profiles, Favorite Albums, Artists and Biographies, Making the Platform More Engaging and Socially Interactive Than a Standard User Account (Musicboard, 2026a).

How The App Was Implemented

From My Current Knowledge and Understanding of This App Was Implemented Using the Following Tech Stack:

Android Component	Purpose
Jetpack Compose	Builds Modern User Interface (Android Developers, 2026a).
Firebase Firestore	Stores Reviews, Ratings and Comments
Firebase Authentication	User Accounts

Cloud Storage	Stores Profile Pictures
RecyclerView/LazyColumn	Displays Review Feeds (Android Developers, 2026b).
Navigation Component	Screen Navigation (Android Developers, 2026c).
Retrofit	Retrieves Album Information from Music Database (Android Developers, 2026d).
Paging Library	Efficiently Loads Long Review Feeds (Android Developers, 2026e).
Push Notifications	Alerts Users to New Album Releases
Material Design 3	Modern Android Interface (Google, 2026).

A Typical Workflow Would Be:

1. User Searches for An Album.
2. Album Metadata Is Retrieved from A Music Database API (Android Developers, 2026d).
3. User Submits a Rating and Review.
4. Review Is Stored in Firebase Firestore
5. Followers Receive Updates
6. Rankings And Recommendations Are Generated from Community Ratings (Musicboard, 2026a).

App Screenshots (Musicboard)

Register/Sign-Up Screen

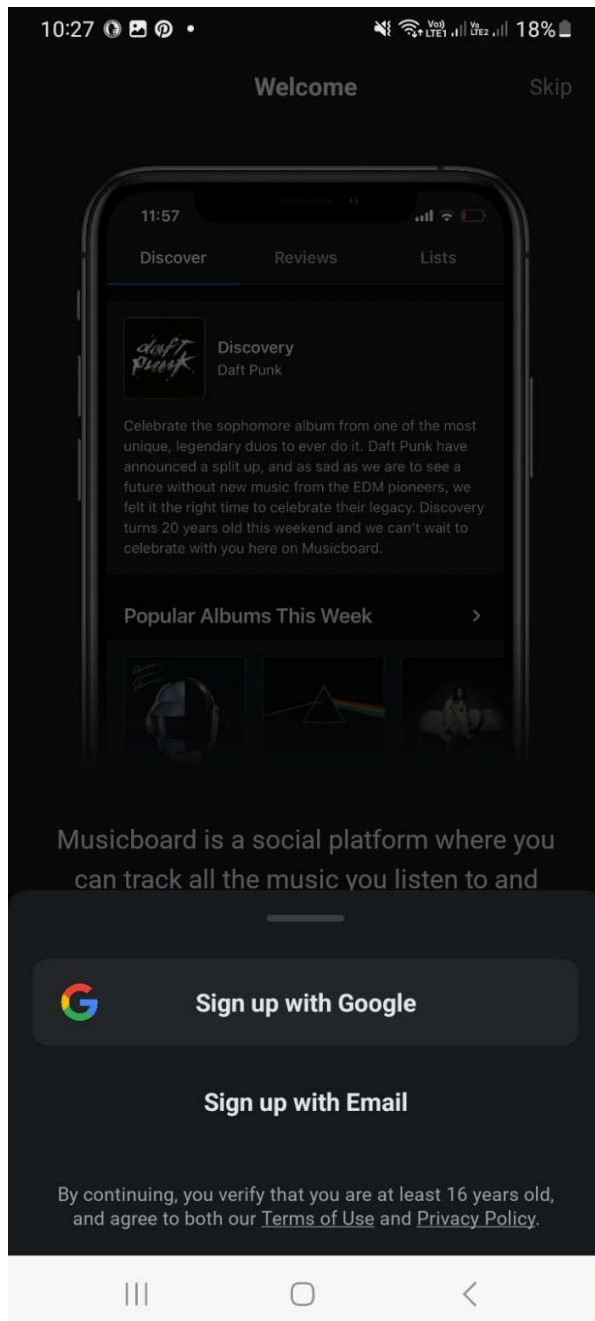


Figure 7: Musicboard Register/Sign-Up Screen

Source: Musicboard (2026a).

This Is the Sign-Up Page of Musicboard, This Allows the User to Create an Account If They Are Using the Application for The First Time. As Seen Above They Are Able to Sign Up Using Their Google Account or Their Email Address.

Home Screen

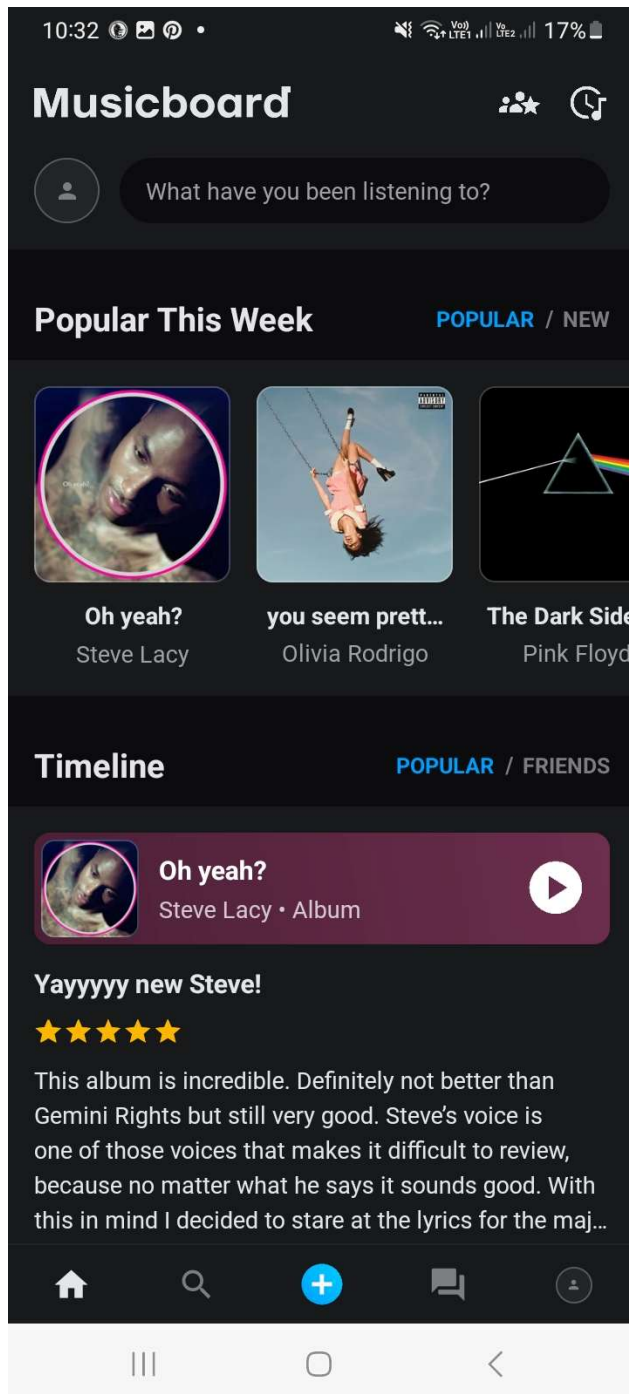


Figure 8: Musicboard Home Screen

Source: Musicboard (2026a).

This Is the Application's Homepage, It Acts as A Hub of Sorts That Allows the User to View Reviews Published by Other Users of The Application About Albums and Artists. Gives Real-Time Updates About the Most Popular Artist and Album's Currently and Has a Shortcut for The Search Bar Which Allows the User to Search for A Specific Artist or Album.

Album Screen

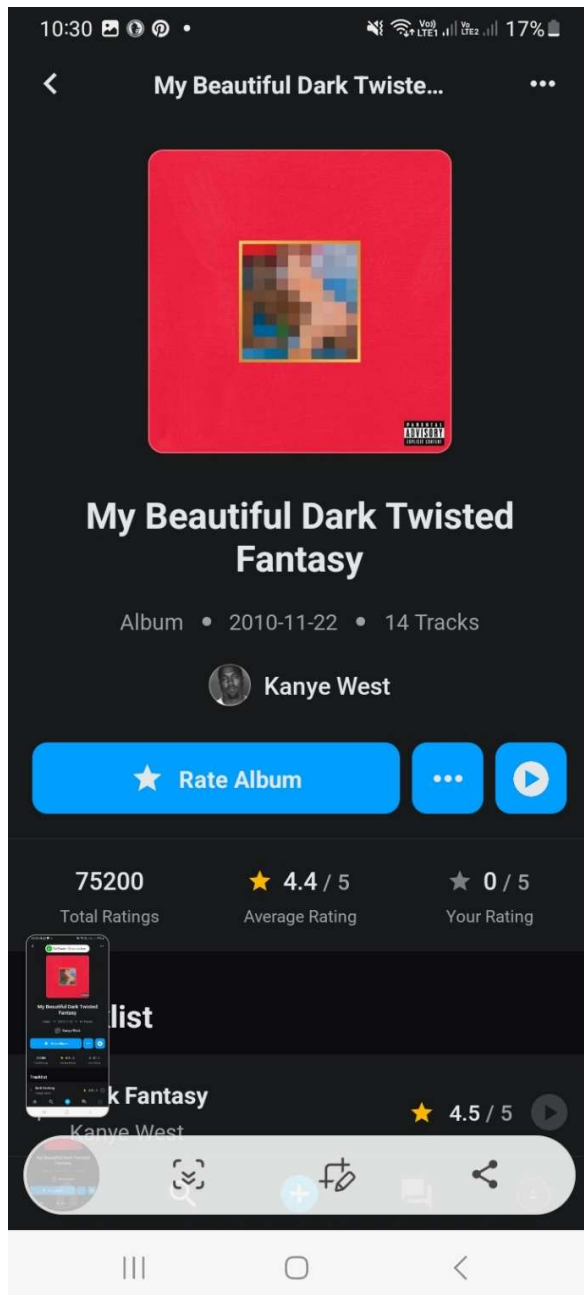


Figure 9: Musicboard Album Screen

Source: Musicboard (2026a).

This is where users are able to view albums released by any artist of their choice. In this case, it is 'My Beautiful Dark Twisted Fantasy' by Kanye West. From here, they are able to view various information and details about the album, such as the date of release, the number of tracks, the total ratings, the average

Rating and The User's Own Rating. The User Has the Option to Rate the Album Using a Four-Star Rating System and They Can Also Play the Album as Well.

Profile Screen

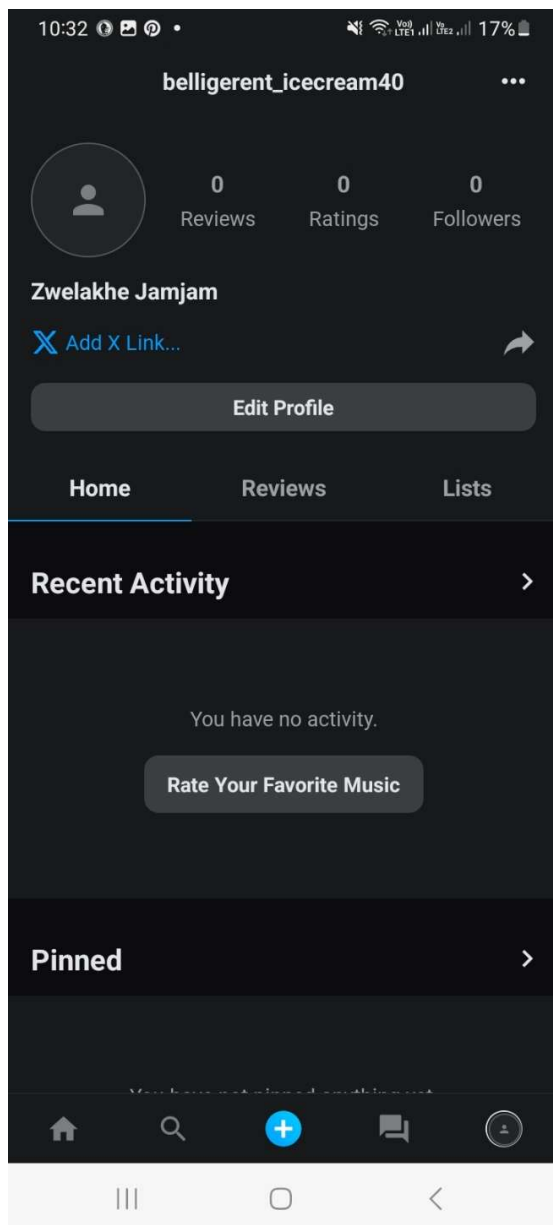


Figure 10: Musicboard Profile Screen

Source: Musicboard (2026a).

This Is the Profile Page, In This Page the User Is Able to View Their Current Profile and Information Related to It Such as Reviews, Ratings, Followers and They Are Also Able to Edit Their Profile's Information Such as Their Username, Add Links, etc. It Also Shows the User's Recent Activity Such as If They Rated Any Albums, It Also Enables Users to View Reviews That They Might Have Made as Well as Lists and View Any Saved Pins That They Have Made.

Overview Of Spotify



Figure 11: Spotify Logo

Source: Spotify (2026a).

Spotify is a global market leader in audio streaming, providing millions of active users with instant on-demand access to a vast media catalog spanning songs, podcasts, and audiobooks. Beyond serving as a simple media delivery network, Spotify's competitive advantage relies heavily on complex data science, machine learning models, and cross-hardware integration designed to maximize user retention and continuous engagement.

Strengths And Weaknesses of Spotify

Strengths	Weaknesses
Massive cloud-based music catalogue with high-fidelity audio playback.	Memory bloats from local bitmap image caching and audio pre-buffering (Spotify, 2026a).

State-of-the-art recommendation engine driven by machine learning algorithms.	Lack of granular album review or community scoring tools (restricted primarily to binary "liking" tracks) (Spotify, 2026b).
Multi-device synchronization and offline playlist downloads for premium users.	Audio ads and playback restrictions for non-subscription users (Spotify, 2026c).
Fluid, dark-themed UI featuring real-time playback transitions and animated album art.	High battery and cellular data consumption due to continuous cloud streaming and animated background visuals (Spotify, 2026d).

Why These Strengths Are Considered Are Innovative

- Massive Cloud-Based Music Catalogue with High-Fidelity Audio Playback:** Instead of requiring users to store physical or digital audio files locally, Spotify provides instant, on-demand cloud streaming to tens of millions of tracks. This transformed music consumption from a file-ownership model into an accessible service, allowing users to discover both global hits and obscure indie tracks instantaneously.
- State-of-the-Art Recommendation Engine (Machine Learning):** Spotify leverages advanced machine learning, collaborative filtering, and natural language processing to analyze individual listening habits and audio features (such as tempo, key, and timbre). This allows features like *Discover Weekly* and *Release Radar* to deliver highly accurate, personalized music suggestions rather than generic top-chart recommendations.
- Multi-Device Synchronization & Offline Downloads:** Through technology like Spotify Connect, users can seamlessly transfer active audio playback across devices (e.g., from smartphone to smart TV or PC) in real time without interrupting the stream. Coupled with encrypted local caching for offline listening, it ensures an uninterrupted user experience regardless of network connectivity.
- Fluid, Dark-Themed UI with Animated Album Art:** Spotify introduced dynamic visual transitions, real-time lyric synchronization, and looping video album art (Canvas). By dynamically extracting color palettes directly from album covers to tint UI components, it creates an immersive, visual-first media experience optimized for modern mobile OLED screens.

How The App Was Implemented

Spotify's Android engineering leverages advanced Jetpack components and high-performance native audio tools:

- **Architecture:** Uses **MVVM (Model-View-View Model)** or **MVI (Model-View-Intent)** with **Kotlin Coroutines and Flow** to manage asynchronous data streams safely (Android Developers, 2026a).
- **Audio Engine:** Utilizes **AndroidX Media3 (ExoPlayer)** operating within a background **Foreground Service** (MediaSessionService). This maintains playback continuity when the app is minimized or the screen is off, while integrating directly with lock-screen controls via the **MediaSession API** (Android Developers, 2026a).
- **UI Layer:** Developed using **Jetpack Compose** alongside custom XML layouts. Dynamic background colour transitions are computed asynchronously using the **Palette API** to sample colours from album cover Bitmaps loaded via image-loading libraries like **Coil** or **Glide** (Android Developers, 2026a).
- **Networking & Data:** **Retrofit** and **OkHttp** handle RESTful and WebSocket API endpoints. Local caching is backed by **Room Persistence Library** or encrypted local file storage (`Context.getExternalFilesDir()`) (Google, 2026).

App Screenshots (Spotify)

Search Screen

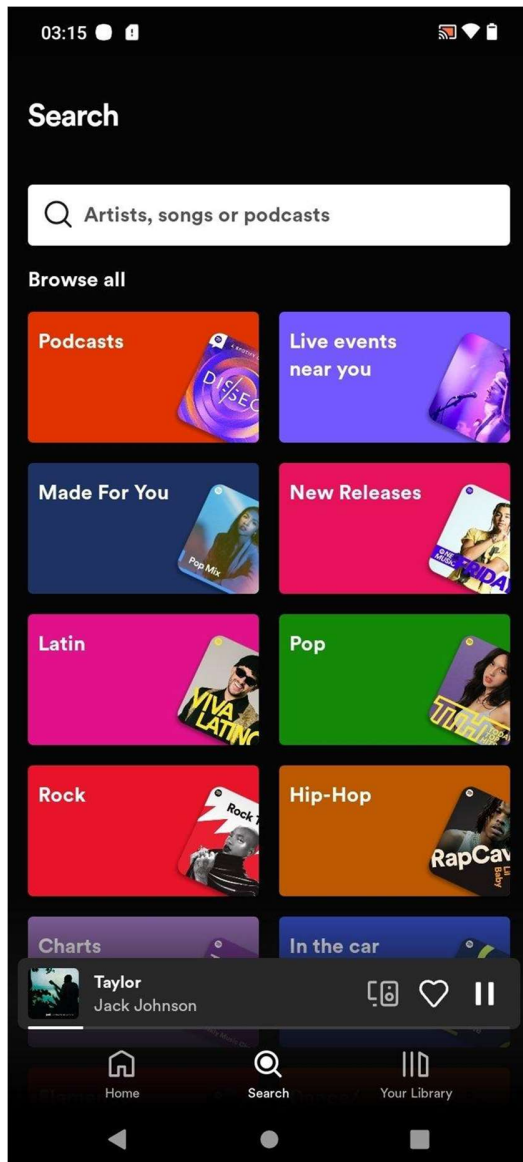


Figure 12: Spotify Search Screen

Source: Spotify (2026a).

Screen Description: Illustrates the main search screen interface with global query capabilities and genre category cards.

Key Components:

- **Top Input Bar:** Search bar prompting "Artists, songs or podcasts".
- **Browse Grid:** Vibrant, color-coded visual cards with tilted cover art for exploring content categories (*Podcasts, Live events, Made for You, New Releases, Latin, Pop, Rock, Hip-Hop, Charts, In the car*).
- **Bottom Navigation:** Persistent bottom player bar playing *Taylor* by Jack Johnson above the standard Android/Spotify navigation controls (*Home, Search, Your Library*).

Home Page

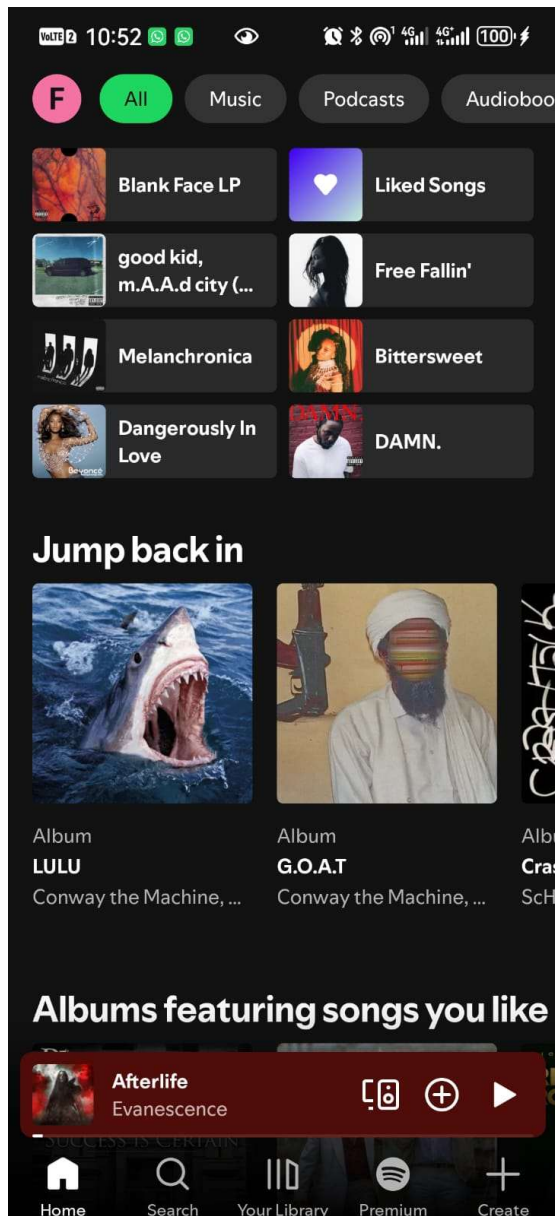


Figure 13: Spotify Home Screen

Source: Spotify (2026a).

Screen Description: Demonstrates the main Spotify home feed with top content filter pills (All, Music, Podcasts, Audiobooks).

Key Components:

- **Top Grid:** Quick access cards for recent/liked playlists (e.g., *Liked Songs*, *good kid*, *m.A.A.d city*, *DAMN.*, *Blank Face LP*).
- **Content Sections:** Categorized horizontal scrolling rows like "*Jump back in*" displaying recently played albums (*LULU*, *G.O.A.T* by Conway the Machine) and "*Albums featuring songs you like*".
- **Bottom Navigation:** Persistent mini player showing currently playing track (*Afterlife* by Evanescence) alongside bottom tab bar navigation (*Home*, *Search*, *Your Library*, *Premium*, *Create*).

Now Playing Screen

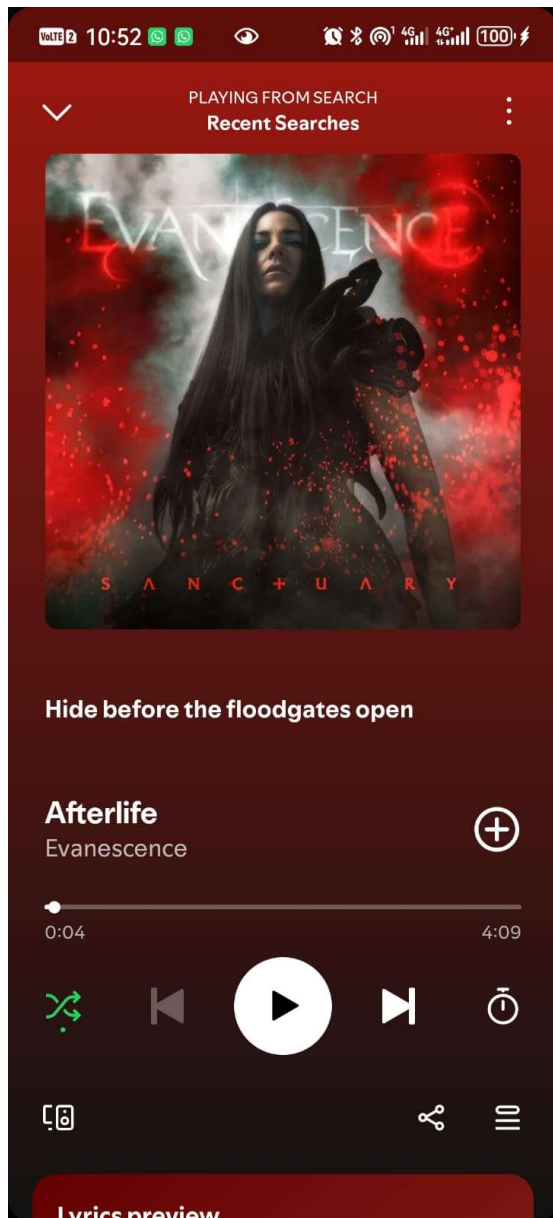


Figure 14: Spotify Now Playing Screen

Source: Spotify (2026a).

Screen Description: Displays Spotify's full-screen audio player interface featuring dark red background tinting extracted from the active album cover.

Key Components:

- **Features the prominent album cover art:** (*Sanctuary / Afterlife* by Evanescence),
- **real-time synchronized lyric snippet:** ("Hide before the floodgates open"),
- **track timeline scrub bar**
- **media control playback buttons:** (Play/Pause, Skip, Shuffle, Repeat), and
- **quick action icons:** (Add to Library +, Connect to Device, Share, Queue).

User Profile Screen

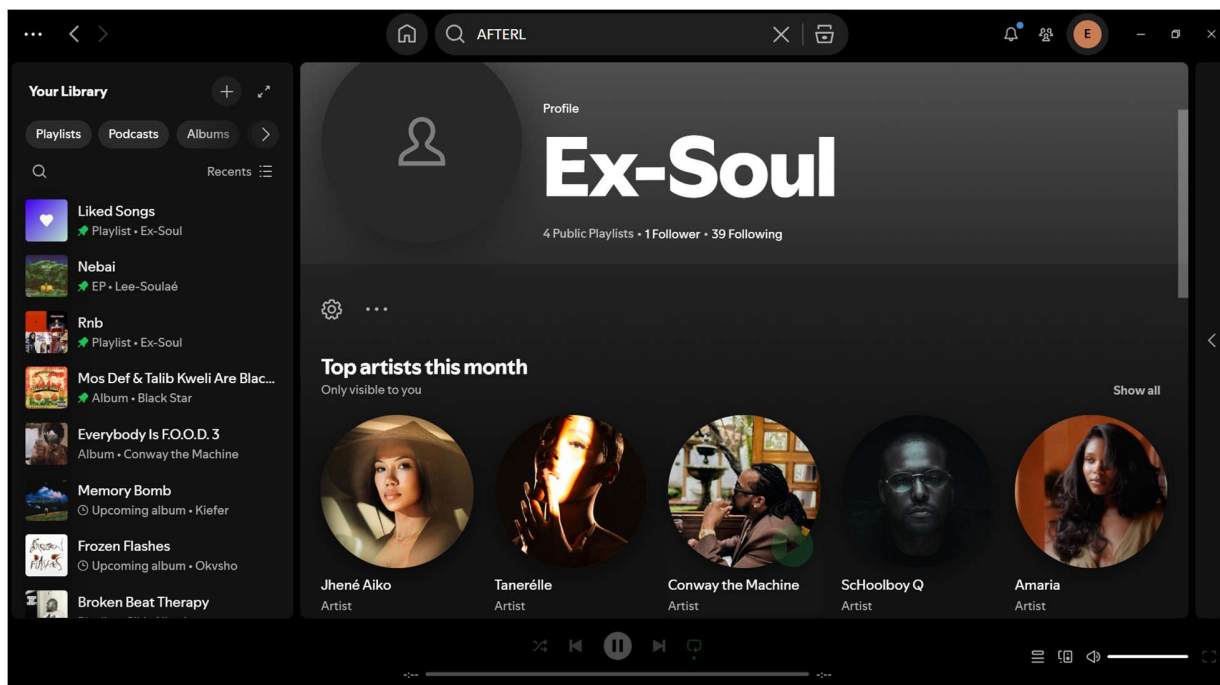


Figure 15: Spotify User Profile Screen

Source: Spotify (2026a).

Screen Description: Displays the Spotify Desktop / Web Application user profile interface for the account "Ex-Soul", showcasing personalized listening analytics and side-bar library navigation.

Key Components:

- **Header & Search Bar:** Displays top search query input (AFTERL), profile avatar navigation, and platform notification shortcuts.
- **User Profile Header:** Features the user display name (**Ex-Soul**), profile avatar placeholder, and social connections counter (*4 Public Playlists • 1 Follower • 39 Following*).
- **Top Artists Section:** A private analytics showcase labeled "*Top artists this month*" displaying circular artist avatars for **Jhené Aiko**, **Taneréle**, **Conway the Machine**, **Schoolboy Q**, and **Amaria**.
- **Left Navigation Drawer (Your Library):** Contains content filters (Playlists, Podcasts, Albums) alongside saved releases and upcoming albums (e.g., *Liked Songs*, *Nebaii*, *Rnb*, *Black Star*, *Everybody Is F.O.O.D. 3*, *Memory Bomb*).
- **Bottom Audio Controls:** Features full media playback functionality, including timeline progress bar, track controls (Play/Pause, Shuffle, Repeat), volume slider, and active device selection buttons

Comparison Of All Three Apps

 LAST.FM vs MUSICBOARD vs SPOTIFY  THE ULTIMATE MUSIC PLATFORM COMPARISON			
Platform	 last.fm Track. Analyse. Discover.	 musicboard Letterbox for Music.	 Spotify Listen. Discover. Everywhere.
 PRIMARY PURPOSE	Track your listening across platforms and discover music through data & analytics.	Rate, review and discuss albums with a social community of music lovers.	Stream music, discover new music and enjoy personalised recommendations.
 LISTENING TRACKING	✓ Automatic scrobbling from streaming services, apps & devices.	✗ No automatic tracking. Users add and rate music manually.	✓ Tracks listening within the app. No tracking outside Spotify.
 MUSIC STREAMING	✗ No music streaming. Works with other streaming services.	✗ No music streaming. Focuses on reviews and community.	✓ Full music streaming service with millions of songs and podcasts.
 REVIEWS & RATINGS	★ Basic ratings (1-5) for tracks, artists & albums. Limited reviews.	✓ Core feature. In-depth album reviews and ratings from the community.	✓ Users can like songs, create playlists and follow reviews (limited).
 MUSIC DATABASE	📦 Large database focused on artists, tracks, albums & listening data.	📦 Large album-focused database with rich metadata.	📦 Huge library with songs, albums, podcasts, artists & playlists.
 DISCOVERY APPROACH	🔍 Algorithm & data-driven recommendations based on listening history.	👥 Community-driven discovery through reviews, lists and social activity.	🌟 Algorithmic + editorial playlists, Discover Weekly, Release Radar & more.
 COMMUNITY & SOCIAL	👤 Following friends, charts, shouts. Community features reduced over time.	👥 Strong social feed, follows, comments, reviews and community interaction.	👤 Follow friends & artists, share playlists, see what friends are listening to.
 CUSTOM LISTS & COLLECTIONS	📋 Limited to playlists and library.	📋 Custom lists for albums, top albums, wishlists, rankings and more.	📋 Playlists, Liked Songs, folders, collaborative playlists.
 COMPATIBILITY & INTEGRATIONS	🔗 Works with Spotify, Apple Music, YouTube Music, Tidal & many others for scrobbling.	🔗 Integrates with music databases & platforms for album data.	🔗 Available on all major platforms and devices. Connects with many apps.
 BEST FOR	🏆 People who want automatic tracking, stats and personalised recommendations.	🏆 People who love reviewing, rating and discovering music through communities.	🏆 People who want to stream music, discover new songs and create playlists.
 PLATFORMS	📱 Web, Android, iOS, Windows, macOS, Android Auto, Apple Watch	📱 Web, Android, iOS	📱 Web, Android, iOS, Windows, macOS, Smart TVs, Consoles, Wearables
 PRICING MODEL	💎 Free with ads & limited features. Last.fm Pro offers advanced stats & customization.	💎 Free with ads & limited stats. Musicboard Pro unlocks advanced insights.	💎 Free (with ads, limited skips). Premium offers ad-free listening & more features.
LAST.FM ✓ Automatic scrobbling ✓ Detailed listening statistics & reports ✓ Personalised recommendations ✓ Similar artists & tracks ✓ Listening compatibility ✓ Cross-platform tracking ✓ Weekly & yearly listening charts KEY FEATURES AT A GLANCE MUSICBOARD ✓ Album reviews & ratings ✓ Detailed written reviews ✓ Custom lists & rankings ✓ Social feed & community ✓ New release tracking ✓ Profile customization ✓ Review notifications SPOTIFY ✓ Millions of songs & podcasts ✓ High-quality music streaming ✓ Personalised playlists (Discover Weekly, etc.) ✓ Offline listening ✓ Cross-device sync ✓ Collaborative playlists ✓ Podcasts & exclusive content			
SIMILARITIES 🎵 All three help users discover new music. ★ All have rating or like systems. 👥 All include community or social features. 📊 All offer personalised recommendations. 📦 All maintain large databases of music and metadata.			
 THE BOTTOM LINE  Choose Last.fm if you want automatic tracking, detailed stats and analytics.  Choose Musicboard if you love reviews, community discussions and album ratings.  Choose Spotify if you want to stream music, create playlists and listen anywhere.			

Figure 16: Comparison Infographic

Source: Developed by the author (2026).

Features We Would Like to Implement in Our App

Last.fm

- **Similar Artist Recommendations:** Many Music Listeners Often Struggle to Discover Artists Beyond Mainstream Recommendations. Similar Artists Recommendations Solve This Problem by Introducing Users to Musicians They Are Likely to Enjoy Without Requiring Them to Search Manually. Unlike Generic Recommendation Systems That Simply Display Popular Music, This Feature Analyzes Listening Behavior and Musical Relationships to Provide Personalized Suggestions. This Is Innovative Because the Recommendation Engine Uses User Behavior and Similarity Algorithms Rather Than Simple Keyword Searches, Making Music Discovery Intelligent Rather Than Random (Last.fm, 2026a).
- **Personal Listening Statistics:** People Enjoy Learning More About Their Habits. Listening Statistics Transform Ordinary Listening History into Meaningful Insights That Users Can Explore and Share. Features Such as Spotify Wrapped Has Demonstrated the Popularity of Listening Analytics. This Is Innovative Because Rather Than Simply Storing Listening History, The Application Transforms Raw Data into Visual Analytics (Last.fm, 2026b).

Musicboard

- **Album Ratings:** Ratings Allow Users to Quickly Express Opinions Without Writing Lengthy Reviews. The Ratings Also Contribute to Community Rankings and Recommendations Algorithms. This Is Innovative Collective Ratings Create a Community-Driven Recommendation System That Becomes More Accurate as More Users Participate (Musicboard, 2026a).
- **Album Rankings:** Rankings Encourage Users to Organize Their Music Collections While Sharing Personal Tastes with Others. They Also Create Opportunities for Discussion. This Is Innovative Because Instead of Fixed Playlists, Rankings Reflect User Preferences and Change Over Time (Musicboard, 2026b).

- **Profile Customization:** Personalization Creates a Stronger Emotional Connection Between Users and The Application. Profiles Become an Expression of Musical Identity. This Is Innovative Because Profiles Become Digital Music Identities Instead of Simple User Accounts (Musicboard, 2026c).
- **Community Recommendations:** Recommendations from Real People Are Often Viewed as More Trustworthy Than Purely Algorithmic Suggestions. Users Discover Music Through Shared Experiences and Discussions. This Is Innovative Combines Social Networking with Recommendation Systems, Creating A Human-Centered Approach (Musicboard, 2026d).
- **New Release Tracking:** Many Users Miss New Release Because They Follow Numerous Artists Across Multiple Platforms and Automatic Notification Ensure Users Never Miss New Music. This Is Innovative Because It Transforms the Application into An Active Music Companion Rather Than a Passive Music Database (Musicboard, 2026e).
- **Custom Music Lists:** Unlike Playlists, Custom Lists Organize Music Around Themes, Opinions or Personal Interests, Allowing Users to Curate Collections That Reflect Their Tastes and Experiences. This Is Innovative Because Custom Lists Go Beyond Playback by Enabling Users to Create and Share Curated Collections That Can Inspire Other Listeners (Musicboard, 2026f).
- **Written Music Reviews:** Reviews Create Meaningful Discussion and Provide Context That Numerical Ratings Alone Cannot Capture. They Help Users Decide Whether an Album Aligns with Their Musical Tastes While Fostering a Knowledgeable and Engaged Community. This Is Innovative Because Unlike Simple Star Ratings, Written Reviews Capture Qualitative Experiences and Personal Perspectives, Transforming the Application into A Platform for Music Criticism and Community Engagement (Musicboard, 2026g).

Spotify

- **In-App 30-Second Audio Previews:** Integrates an embedded audio player into album and review screens, letting users preview tracks instantly while reading community feedback. This Bridges the gap between static text reviews and audio playback, allowing users to immediately audition music without leaving the app (Spotify, 2026a).

- **Spotify Web API Instant Search:** Connects the search bar to a global music database API, offering real-time auto-complete for artists, albums, and tracks. This Eliminates the need to maintain a costly manual metadata database by leveraging cloud-based API endpoints for up-to-date music metadata and high-resolution cover art (Spotify, 2026b).
- **Dynamic Palette Tinting (UI/UX):** Dynamically extracts dominant colors from album artwork to tint background gradients and card UI components in real time. This Delivers a modern, visually immersive dark-mode interface that adapts visually to whichever album or track the user is currently inspecting (Spotify, 2026c).

Conclusion

This Research Document Has Provides a Comprehensive Analysis of Three Established Music Applications; Last.fm, Musicboard and Rate Your Music to Gain a Deeper Understanding of The Features Design Principles and Technologies Commonly Used in Modern Music Discovery Platforms. By Evaluating The Strengths and Weaknesses of Each Application, Comparing Their Functionality, and Identifying Their Most Valuable Features. This Report Has Established a Solid Foundation for The Development of Our Own Music Discovery, Tracking and Recommendation Application.

The Comparison Demonstrated That Each Application Offers Unique Advantages, Ranging from Personalized Music Recommendation and Listening History Tracking to Community Engagement, Album Reviews and Curated Music Lists. These Findings Have Enabled Us to Identify the Features That Provide the Greatest Value to Users and Determine How They Can Be Combined to Create a More Engaging and User-Friendly Application.

The Outcomes of This Research Will Directly Inform the Planning and Design Phase of The Project. The Selected Features and Insights Gathered Throughout This Report Will Guide the Application's Functional Requirements, User Interface Design, API Integration, Database Structure and Overall System Architecture. By Building Upon Proven Concepts While Introducing Improvements Tailored to Our Application's Objectives, The Design Document Will Establish a Clear and Structured Blueprint for Developing a High-Quality Music Discovery and Recommendation Application That Meets Both User Needs and The Technical Requirements of The Project.

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Write-Up

The AI Tools That Were Used During the Completion of This Assignment Was Piktochart AI Which Converts Documents into Structured, long-Form Infographics and Reports. It Was Used to Create the Infographic Using a Document That Contained All the Information That I Had Written About Last.fm, Musicboard and Raate Your Music. Piktochart AI Was Cited in The References Section of This Document.